

RF-1 AKLT Sandwich

Audit & Crosscheck Report

Full corpus verification of RCF-RES-v6.1-RF1

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Audit Date	2026-08-26
Auditor	Z.ai (automated verification + crosscheck)
Scope	Full corpus audit (RF-1 + lineage v1 to v6 + Peer Review + Variance Retraction + Con.pdf)
Arithmetic	Exact rational (sympy) — 11/11 checks PASSED
Verdict	Math CORRECT ; 6 MAJOR + 5 MINOR + 3 NIT documentation issues

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1. Executive Summary

This report documents a full-corpus audit of Research Increment RF-1 (RCF-RES-v6.1-RF1: "The AKLT Sandwich Resolution") against the existing RCF corpus comprising V6_0, V4_0, v3.0 Canonical Rewrite, v2.1 Patch Plan, v1-to-v2 Change Log, Critical Peer Review (v2.1 vs v3.0), the Variance Retraction Addendum, and the standalone manuscript Con.pdf. The audit combines (i) exact rational arithmetic verification of all numerical claims in RF-1 Theorem RF-1.1 using a standalone sympy script, and (ii) a cross-version consistency pass over the lineage v1 to v2 to v2.1 to v3 to V4 to V6.

Mathematical verification verdict: PASS. All 11 numerical claims in RF-1 Theorem RF-1.1 are confirmed by exact rational diagonalization of the 81×81 sandwich operator $S = P_{\text{odd}} P_{\text{even}} P_{\text{odd}} (1 - P_K)$ on the $L = 4$ AKLT spin-1 chain. The spectrum, the maximizing state $|\psi_{\text{top}}\rangle = |\varphi_{12} \otimes \varphi_{34}$ with $|\varphi\rangle = |1,0\rangle - |0,1\rangle$, the $B*B$ identity consequence, the ground-space orthogonality, and the Detectability Lemma bound $\Delta_{\text{DL}} = 1/6$ are all reproduced exactly. The verification script is delivered as a standalone artifact at `/home/z/my-project/download/aklt_verify.py` and can be re-executed at any time.

Documentation / provenance / lineage verdict: 6 MAJOR, 5 MINOR, and 3 NIT issues identified. The math is correct, but RF-1 carries six substantive documentation issues that should be fixed before the increment is integrated into v6.2: (P1) class/status conflation in "Class IV / E [Established]" notation; (P2) mainline truncation that drops V6_0's ε -tail; (P3) DL-term reinsertion that reverses V6_0's DL-13 gate demotion; (P4) "every term derived from closed-form algebra" wording that understates Δ_{∞} 's literature-derived Class L provenance; (P5) $P_{\text{odd}} / P_{\text{even}}$ definitions that are new definitional content, not consolidation; (P6) §6.2 numbering collision between V4_0 (Quantum Darwinism) and V6_0 (AKLT chain). Five additional MINOR issues (P7-P11) and three NITs (P12-P14) round out the patch list. None of these are mathematical blockers.

Recommended action. Apply patches P1-P6 (MAJOR) before promoting RF-1 to v6.2; batch P7-P11 (MINOR) in a follow-up sweep; P12-P14 (NIT) are optional. The redlined patch document is delivered alongside this report at `/home/z/my-project/download/RCF-RES-v6.1-RF1-audit-patch.md`.

Check	Claim	Status
C1	$\dim H = 81$	PASS
C2	$\dim K = 4$ (ground space)	PASS
C3	J-sector decomposition $(3,6,6,3,1) \times (1,3,5,7,9) = 81$	PASS
C4	Spectrum $\{0, 1/36, 1/6, 11/36, 1/3\}$ matches §2	PASS
C5	$\ S\ _{\infty} = 1/3$ (multiplicity 5)	PASS
C6	$\ B\ _{\infty} = 1/\sqrt{3}$ via $B*B = S$ identity	PASS
C7	$ \psi_{\text{top}}\rangle$ is eigenvector with eigenvalue $1/3$	PASS
C8	$P_K \psi_{\text{top}}\rangle = 0$ (orthogonality)	PASS
C9	$\Pi_{\{23\}}$ action on two-site basis matches §3.3	PASS
C10	Arithmetic $(1/8 + 1/12 + 1/8) = 1/3$	PASS
C11	$\Delta_{\text{DL}} = (1 - \gamma^2)/g^2 = 1/6$	PASS

Table 1.1: Numerical verification verdict (11/11 PASS).

2. Verification Methodology

2.1 Exact Rational Arithmetic Pipeline

The mathematical verification was performed using a standalone Python script (`/home/z/my-project/download/aklt_verify.py`) that constructs the $L = 4$ AKLT spin-1 chain Hilbert space $H = (C^3)^{\otimes 4}$ of dimension 81 explicitly, lifts the two-site spin-2 bond projector $\Pi = \sum_{\{M=-2\}^2} |2, M\rangle\langle 2, M|$ to the full 81×81 space via Kronecker embeddings, builds the layer projectors $P_{\text{odd}} = (1 - \Pi_{12})(1 - \Pi_{34})$ and $P_{\text{even}} = 1 - \Pi_{23}$, computes the ground space $K = \ker(M_{\text{hat}})$ where $M_{\text{hat}} = \Pi_{12} + \Pi_{23} + \Pi_{34}$ is the AKLT Hamiltonian, constructs the orthogonal projector P_K onto K , forms the sandwich operator $S = P_{\text{odd}} P_{\text{even}} P_{\text{odd}} (1 - P_K)$, and then diagonalizes S exactly using sympy Rational arithmetic.

No floating point arithmetic is used in any of the core linear-algebra operations. All eigenvalues are returned as exact sympy Rational numbers (e.g., `Rational(1, 36)`, `Rational(11, 36)`, `Rational(1, 3)`), so the spectrum is provably exact — not merely a float approximation that matches to $1e-12$. The script also constructs the J_{tot}^2 operator via $J_{\text{tot}} = \sum_i S_i$ (where S_i are the spin-1 single-site operators lifted to the full 81-dim space) and diagonalizes it to verify the J-sector decomposition independently of the sandwich spectrum. The two computations are mutually independent and confirm RF-1's claims from two directions.

2.2 Cross-Version Lineage Audit

The corpus crosscheck was performed by four parallel reading agents, each consuming a different subset of the corpus:

- **Agent A:** `V6_0.pdf` (the baseline RF-1 extends) — extracted status taxonomy, VN-17/VN-18 entries, §6.2(b) mainline, and AKLT chain definitions.
- **Agent B:** `v3.0 Canonical Rewrite + Variance Retraction Addendum` — extracted earlier-version status taxonomy, confirmed VN-17/VN-18 are V6_0-introduced (no v3.0 precedent), and confirmed the Variance Retraction is orthogonal to the AKLT/DL-bound content.
- **Agent C:** `Con.pdf` (largest file, 462 KB) — identified `Con.pdf` as a standalone manuscript (NOT a compendium), with a simpler 4-category status taxonomy and no AKLT/DL-bound content. `Con.pdf` is orthogonal to the RF-1 audit; it represents an earlier or parallel lineage of the Relational Constraint Framework.
- **Agent D:** `V4_0.pdf + v1-to-v2 Change Log + v2-to-v2.1 Patch Plan + Critical Peer Review (v2.1 vs v3.0)` — established the version chain `v1→v2→v2.1→v3→V4→V6`, confirmed VN-17/VN-18 are absent in V4_0 and earlier, identified the §6.2 numbering collision between V4_0 and V6_0, and flagged two unresolved Peer Review concerns that RF-1 inadvertently worsens.

Each agent was given a structured prompt requesting specific extracts (status taxonomy definitions, VN-XX entries, §6.2(b) mainline ordering, $P_{\text{odd}}/P_{\text{even}}$ definitions, sandwich operator references, J-sector decomposition, bridging identity, constants values, AALV/C-02 citation status). All four agents returned comprehensive findings; their full reports are archived in `/home/z/my-project/worklog.md` (shared multi-agent worklog).

3. Numerical Verification Results

This section presents the exact numerical verification of RF-1 Theorem RF-1.1. All 11 checks passed; the spectrum, the maximizing state, the B^*B identity, the ground-space orthogonality, and the Detectability Lemma bound are all reproduced exactly. The script `aklt_verify.py` produces the raw output log delivered at `/home/z/my-project/download/aklt_verify_output.txt`.

3.1 Sanity Checks (Single-Site and Two-Site Operators)

Before building the full 81×81 operators, the script performs two sanity checks on smaller subsystems. First, the two-site spin-2 projector $\Pi = \sum_{M=-2}^2 |2, M\rangle\langle 2, M|$ is constructed on the 9-dimensional two-site Hilbert space using Clebsch-Gordan coefficients for the coupling $1 \otimes 1 = 0 \oplus 1 \oplus 2$. The eigenvalues are $\{0$ (multiplicity 4), 1 (multiplicity 5) $\}$, matching the decomposition: the spin-2 subspace is 5-dimensional (containing the $J = 2$ multiplet with $2J+1 = 5$ states), and the spin-0/spin-1 subspace together is 4-dimensional (1 state for $J = 0$ and 3 states for $J = 1$). The projector is idempotent ($\Pi^2 = \Pi$) and self-adjoint ($\Pi^\dagger = \Pi$), as required.

Second, the single-site spin-1 operators S_x, S_y, S_z are constructed in the standard basis $|+1\rangle, |0\rangle, |-1\rangle$ using the raising operator S_+ with matrix elements $\langle 1|S_+|0\rangle = \sqrt{2}$ and $\langle 0|S_+|-1\rangle = \sqrt{2}$. The sanity check verifies that $S \cdot S = S_x^2 + S_y^2 + S_z^2 = 2 \cdot I_3$, matching the expected $S(S+1) = 1(1+1) = 2$ for spin-1. This confirms the spin operators are correctly constructed before they are lifted to the full 81-dim space for the J_{tot}^2 calculation in §3.5 below.

3.2 Full Spectrum of the Sandwich Operator S

The sandwich operator $S = P_{\text{odd}} P_{\text{even}} P_{\text{odd}} (1 - P_K)$ is constructed on the 81-dimensional Hilbert space by lifting the 9×9 two-site Π to the full space via Kronecker products with identity on the spectator sites. The bond projectors $\Pi_{12}, \Pi_{23}, \Pi_{34}$ are all idempotent and self-adjoint, confirming they are valid orthogonal projectors. The layer projectors $P_{\text{odd}} = (1 - \Pi_{12})(1 - \Pi_{34})$ and $P_{\text{even}} = 1 - \Pi_{23}$ are then constructed, and the AKLT Hamiltonian $M_{\text{hat}} = \Pi_{12} + \Pi_{23} + \Pi_{34}$ is diagonalized to find its kernel (the ground space K). The kernel has dimension 4 ($\dim K = 4$), matching RF-1 §1 and V6_0 §6.2(b) page 35 line 1440: " $\ker(M_{\text{hat_AKLT}}) = \cap_j \ker(\Pi_j)$ is the valence-bond MPS ground space: $\dim K = 4$ open boundary".

The orthogonal projector P_K onto K is constructed via $P_K = B(B^T B)^{-1} B^T$ where B is the 81×4 matrix whose columns are a basis of K . P_K is verified idempotent ($P_K^2 = P_K$) and self-adjoint ($P_K^\dagger = P_K$), with rank 4. The sandwich operator $S = P_{\text{odd}} P_{\text{even}} P_{\text{odd}} (1 - P_K)$ is then constructed and verified self-adjoint.

Exact diagonalization of S yields the following complete spectrum, matching RF-1 §2 Table exactly:

Eigenvalue	Multiplicity	Float value	RF-1 §2 claim	Status
0	69	0.000000	69 (= 81 – 12 nonzero)	PASS
1/36	1	0.027778	1 ($J = 0$ sector)	PASS
1/6	3	0.166667	3 ($J = 1$ sector $\times$ mult)	PASS
11/36	3	0.305556	3 ($J = 1$ sector $\times$ mult)	PASS
1/3	5	0.333333	5 ($J = 2$ sector $\times$ mult) — λ_{max}	PASS
Total	81	—	Sum of multiplicities	PASS

Table 3.1: Exact rational spectrum of $S = P_{\text{odd}} P_{\text{even}} P_{\text{odd}} (1 - P_K)$.

The maximum eigenvalue is $\lambda_{\max} = 1/3$ with multiplicity 5, matching RF-1 §2 final row (" $\lambda_{\max} = 1/3$ (multiplicity 5)"). This confirms the operator norm $\|S\|_{\infty} = 1/3$ (RF-1 Theorem RF-1.1 first boxed equation) and establishes the sandwich-norm half of the closure.

3.3 Maximizing State Computation

RF-1 §3 constructs an explicit maximizing state $|\psi_{\text{top}}\rangle = |\varphi_{12} \otimes \varphi_{34}\rangle$ with $|\varphi\rangle = |1,0\rangle - |0,1\rangle$ (the antisymmetric $J = 1$, $M = 1$ state on a bond). In the 4-site basis, this expands to $|\psi_{\text{top}}\rangle = |1,0,1,0\rangle - |1,0,0,1\rangle - |0,1,1,0\rangle + |0,1,0,1\rangle$. The script constructs this state as a length-81 column vector with nonzero entries ± 1 at indices 10, 12, 28, 30 (corresponding to the four computational basis states above), and the squared norm is verified to be 4 ($\|\psi_{\text{top}}\|^2 = 4$).

Applying S to $|\psi_{\text{top}}\rangle$ yields a vector with exactly four nonzero components, all equal to $\pm 1/3$ times the corresponding component of $|\psi_{\text{top}}\rangle$:

Index	Basis state	$\psi_{\text{top}}[i]$	$(S\psi_{\text{top}})[i]$	Ratio
10	$ 1,0,1,0\rangle$	+1	+1/3	1/3
12	$ 1,0,0,1\rangle$	−1	−1/3	1/3
28	$ 0,1,1,0\rangle$	−1	−1/3	1/3
30	$ 0,1,0,1\rangle$	+1	+1/3	1/3

Table 3.2: Action of S on $|\psi_{\text{top}}\rangle$ — all four ratios equal 1/3.

Therefore $S|\psi_{\text{top}}\rangle = (1/3)|\psi_{\text{top}}\rangle$, confirming $|\psi_{\text{top}}\rangle$ is an eigenvector of S with eigenvalue $1/3$. This is the maximum eigenvalue, so $|\psi_{\text{top}}\rangle$ is a maximizing state. This verifies RF-1 §3.4 final line (" $P_{\text{odd}} P_{\text{even}} P_{\text{odd}} |\psi_{\text{top}}\rangle = (1/3)|\psi_{\text{top}}\rangle$ ") exactly. Note that the $(1/8 + 1/12 + 1/8) = 1/3$ arithmetic in RF-1 §3.4 also checks: $1/8 + 1/12 + 1/8 = (3 + 2 + 3)/24 = 8/24 = 1/3$.

3.4 Ground-Space Orthogonality ($P_K |\psi_{\text{top}}\rangle = 0$)

RF-1 §3.5 argues that $|\psi_{\text{top}}\rangle$ is orthogonal to the ground space K because $|\psi_{\text{top}}\rangle$ has total magnetic quantum number $M = 2$ (sum of single-site m_j : $1+0+1+0 = 2$ for the first basis state, and similarly for the other three terms), whereas the ground space on $L = 4$ open boundary consists of 4 states belonging to sectors $J = 0$ ($M = 0$) and $J = 1$ ($M \in \{-1, 0, +1\}$). Since all ground-space states have $M \leq 1$ and $|\psi_{\text{top}}\rangle$ has $M = 2$, the selection rule on M forces $P_K |\psi_{\text{top}}\rangle = 0$.

The script confirms this directly: $P_K |\psi_{\text{top}}\rangle$ is the zero vector. Consequently $(1 - P_K) |\psi_{\text{top}}\rangle = |\psi_{\text{top}}\rangle$, so the projection onto $(1 - P_K)$ is a no-op for this state, and $S |\psi_{\text{top}}\rangle = P_{\text{odd}} P_{\text{even}} P_{\text{odd}} |\psi_{\text{top}}\rangle = (1/3)|\psi_{\text{top}}\rangle$. This verifies RF-1 §3.5.

3.5 J-Sector Decomposition $(3, 6, 6, 3, 1) \times (1, 3, 5, 7, 9) = 81$

As an independent check (not relying on the sandwich spectrum), the script constructs $J_{\text{tot}} = \sum_j S_j$ (the total angular momentum operator) and diagonalizes $J_{\text{tot}}^2 = J_x^2 + J_y^2 + J_z^2$. The eigenvalues of J_{tot}^2 are $J(J+1)$ for $J = 0, 1, 2, 3, 4$, and the multiplicities give the total dimension of each J -sector. The radial dimensions (number of distinct angular-momentum states per J , ignoring the $(2J+1)$ M -degeneracy) are obtained by dividing the multiplicity by $(2J+1)$.

J	J(J+1)	Total mult	2J+1	Radial dim	RF-1 claim	Status
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0	0	3	1	3	3×1	PASS
1	2	18	3	6	6×3	PASS
2	6	30	5	6	6×5	PASS
3	12	21	7	3	3×7	PASS
4	20	9	9	1	1×9	PASS
Sum	—	81	—	—	$(3,6,6,3,1) \times (1,3,5,7,9)$	PASS

Table 3.3: J-sector decomposition via J_{tot}^2 diagonalization.

The radial dimensions (3, 6, 6, 3, 1) and the $(2J+1)$ multiplicities (1, 3, 5, 7, 9) match RF-1 §2 Table exactly. The sum of multiplicities is $3 + 18 + 30 + 21 + 9 = 81$, matching $\dim H = 81$. This independently confirms RF-1's J-sector decomposition, which was previously established as V6_0 VN-06 (Class IV).

3.6 B^*B Identity and $\gamma_{\text{DL}} = 1/\sqrt{3}$ Consequence

RF-1 Theorem RF-1.1's second boxed equation states that $\gamma_{\text{DL}} := \|P_{\text{even}} P_{\text{odd}} (1 - P_K)\|_{\infty} = 1/\sqrt{3}$, derived from the first equation $\|P_{\text{odd}} P_{\text{even}} P_{\text{odd}} (1 - P_K)\|_{\infty} = 1/3$ by the C^* -algebra identity $\|AB\|^2 = \|BAB\|$. The script verifies this directly: let $B = P_{\text{even}} P_{\text{odd}} (1 - P_K)$. Then $B^*B = (1 - P_K) P_{\text{odd}} P_{\text{even}} P_{\text{even}} P_{\text{odd}} (1 - P_K) = (1 - P_K) P_{\text{odd}} P_{\text{even}} P_{\text{odd}} (1 - P_K)$ (since $P_{\text{even}}^2 = P_{\text{even}}$), which equals $S = P_{\text{odd}} P_{\text{even}} P_{\text{odd}} (1 - P_K)$ because $(1 - P_K)$ commutes with P_{odd} and P_{even} (since P_K is built from M_{hat} which commutes with all bond projectors) and $(1 - P_K)^2 = (1 - P_K)$.

The script confirms $B^*B = S$ exactly (as exact sympy matrices). Therefore $\|B\|_{\infty}^2 = \lambda_{\text{max}}(B^*B) = \lambda_{\text{max}}(S) = 1/3$, and $\|B\|_{\infty} = \sqrt{1/3} = 1/\sqrt{3}$. This confirms RF-1's second boxed equation and the B^*B identity. Note that V6_0 page 53 line 2224 carries the bridging identity $\|PQ(1-K)\|^2 = \|QPQ(1-K)\|$ as the algebraic reduction RF-1 uses; this is the standard C^* -algebra identity, not a novel result of the corpus.

3.7 Detectability Lemma Bound $\Delta_{\text{DL}} = 1/6$

RF-1 §1 "Immediate Baseline Consequence" states that the Detectability Lemma candidate bound $\Delta_{\text{DL}} = (1 - \gamma^2)/g^2 = (1 - 1/3)/2^2 = (2/3)/4 = 1/6$ is grounded on the exhibited certificate. Here $\gamma = 1/\sqrt{3}$ (the sandwich norm) and $g = 2$ (the number of layers in the sandwich, or the interaction strength parameter in the Detectability Lemma framework). The arithmetic is confirmed exactly: $(1 - 1/3)/4 = (2/3)/4 = 1/6$.

This verifies RF-1 §1 Theorem RF-1.1 consequence and grounds VN-18 ($\Delta_{\text{DL}} \geq 1/6$) on an exhibited certificate. The bound is now Established (per V6_0 §5 status vocabulary), conditional on Theorem RF-1.1 itself being Established (which the present verification confirms).

3.8 $\Pi_{\{23\}}$ Action on Two-Site Basis (RF-1 §3.3)

RF-1 §3.3 lists four specific identities for the action of the spin-2 projector $\Pi_{\{23\}}$ on basis states of sites (2, 3):

- $\Pi|1,1\rangle = |1,1\rangle$ (since $|1,1\rangle$ is the $J = 2, M = 2$ state)
- $\Pi|1,0\rangle = (1/2)(|1,0\rangle + |0,1\rangle)$
- $\Pi|0,1\rangle = (1/2)(|1,0\rangle + |0,1\rangle)$
- $\Pi|0,0\rangle = (2/3)|0,0\rangle + (1/3)(|1,-1\rangle + |-1,1\rangle)$

The script verifies each of these by extracting the corresponding column of the 9×9 two-site Π matrix and reading off the nonzero coefficients. All four identities match exactly. This confirms the intermediate step of

RF-1 §3.3, which feeds into the §3.4 sandwich-action computation. The overall maximizing-state computation in §3.4 (which yields $(1/8 + 1/12 + 1/8) = 1/3$) is therefore arithmetically grounded.

4. Cross-Check Against V6_0

V6_0 is the baseline that RF-1 claims to extend. Agent A read V6_0 (~135 KB extracted text, 53 pages) and extracted the status taxonomy, VN-17/VN-18 entries, §6.2(b) mainline ordering, and AKLT chain definitions. This section cross-checks RF-1's claims against V6_0 to identify any drift, conflation, or substantive contradiction.

4.1 Status Taxonomy in V6_0

V6_0 §5 defines a closed set of 8 status tags (separate from VN provenance classes):

- **[Established]**: Theorem with exhibited proof; no open dependencies.
- **[Structural]**: Definitions, postulates, conventions.
- **[Conditional on X]**: Proof exhibited modulo named assumptions.
- **[Closed (scope)]**: Closed with mandatory scope qualifier.
- **[Empirical—Validated]**: Fitted regularity with out-of-sample validation.
- **[Input Verified; Inference OPEN]**: Certificate whose inputs are verified but load-bearing inference unexhibited.
- **[Conjecture / Quarantined]**: Q-register or explicitly flagged.
- **[Retracted]**: Register entries.

V6_0 §6 separately defines 6 VN provenance classes: E (exact), IV (independently verified), C (cited), F (fitted), O (open inference), L (literature-derived benchmark; added by AM-1). These are kept in a separate column from the status tags. RF-1's notation "Class IV / E [Established]" conflates the two columns — "IV / E" is ambiguous (could mean "Class IV or Class E" or "Classes IV and E jointly"), and V6_0's registry has a single Class column per VN entry, not multiple. The audit's proposed patch P1 (see §6 below) is to restate the upgrade as a two-column change: Class O → Class IV (provenance); [Input Verified; Inference OPEN] → [Established] (status).

4.2 VN-17 and VN-18 in V6_0

V6_0's Appendix A (page 43, lines 1828–1832) records VN-17 and VN-18 as follows:

- **VN-17**: $\gamma_{DL} = 1/\sqrt{3}$ | Class **O** | [Input Verified; Inference OPEN] | Input verified (VN-16); norm equality unexhibited | Dependency: ← VN-16
- **VN-18**: $\Delta_{DL} \geq 1/6$ | Class **O** | [Input Verified; Inference OPEN] | Auto-invalidates if VN-17 falsified | Dependency: ← VN-17

V6_0's in-context definition (Chunk F §6.2(b), page 36 lines 1511–1519) elaborates the OPEN status: the transfer-operator input (from VN-16 eigenvalues $\{1, -1/3, -1/3, -1/3\}$) is verified, but the operator-norm equality $\|P_{\text{even}} P_{\text{odd}} (1 - P_K)\|_{\infty} = \sqrt{|\lambda_2|}$ is unexhibited. AM-3 sets up an auto-invalidation rule: VN-18 depends on VN-17; if VN-17 is downgraded or falsified, VN-18 is automatically invalidated.

RF-1's upgrade is therefore precisely targeted: Theorem RF-1.1 exhibits the operator-norm equality $\|P_{\text{even}} P_{\text{odd}} (1 - P_K)\|_{\infty} = 1/\sqrt{3}$ (via $B^*B = S$ and $\|S\|_{\infty} = 1/3$), closing the one specific unexhibited equality that V6_0 carried as Class O. The AM-3 dependency cascade means VN-18 is automatically unblocked once VN-17

is upgraded. The V6_0 baseline explicitly anticipated this upgrade path (page 8 line 306: "May falsify VN-17/18"; page 52 line 2213: "the sandwich computation may falsify VN-17") — so RF-1 is operating precisely within its reserved scope. No scope violation.

4.3 §6.2(b) Mainline Ordering in V6_0 vs RF-1

V6_0 §6.2(b) (Chunk F, page 37 lines 1527–1532) records the mainline as:

$$\Delta_{DL} (OPEN) \geq 1/6 \lesssim \Delta_{Knabe}^{(n=4)} \approx 0.173 \leq \Delta_{Knabe}^{(n=6)} \approx 0.248 \leq E^* \approx 0.296 \leq \Delta_{\infty} \approx 0.350 \leq \varepsilon_6 \approx 0.398 \leq \varepsilon_4 \approx 0.449 \leq \varepsilon_3 = 0.500$$

However, V6_0's final frozen mainline (after the DL-13 gate modification, page 38 lines 1584–1590) is:

$$\Delta_{Knabe}^{(4)} \leq \Delta_{Knabe}^{(6)} \leq E^* \leq \Delta_{\infty} \leq \varepsilon_6 \leq \varepsilon_4 \leq \varepsilon_3$$

with the DL term extracted from the primary chain and presented as a separately-stated OPEN candidate bound immediately above it, and the VN-17/VN-18 dependency stated in prose only. This is a deliberate falsification safety mechanism: by isolating the DL candidate from the certified chain, if VN-17 is later falsified, only the candidate needs revision, not the chain.

RF-1 §4.3 re-inserts the DL term into the main chain as a CLOSED Established bound ($\Delta_{DL} = 1/6$ exactly), reversing DL-13's prose-only demotion, AND truncates the chain at Δ_{∞} , omitting $\varepsilon_6 \leq \varepsilon_4 \leq \varepsilon_3$. The audit's patches P2 and P3 (see §6) address these two issues: P2 restores the ε -tail; P3 explicitly notes the DL-13 reversal and preserves the AM-3 dependency as a lineage annotation. The reversal is logically permitted (RF-1's Theorem RF-1.1 genuinely closes VN-17), but the safety mechanism must be explicitly re-engineered for the new context.

4.4 AKLT Chain and Layer Projectors in V6_0

V6_0 §6.2(b) defines the AKLT chain ($L = 4$, $\dim H = 81$, $M_{\hat{}} = \Pi_{12} + \Pi_{23} + \Pi_{34}$), the bond projectors Π_{12} , Π_{23} , Π_{34} (via VN-22: $\Pi = 1/3 + (1/2)x + (1/6)x^2$ on spin-pair operator $x = S \cdot S$), the kernel projector P_K (via §3 Lemma 3.1.1), and the J-sector decomposition (VN-06: radial dims (3,6,6,3,1), mults (1,3,5,7,9), sum 81). RF-1 inherits all of these unchanged.

However, V6_0 does **NOT** define the layer projectors $P_{\text{odd}} = (1 - \Pi_{12})(1 - \Pi_{34})$ and $P_{\text{even}} = 1 - \Pi_{23}$ algebraically. V6_0 only references the sandwich expression $\|P_{\text{odd}} P_{\text{even}} P_{\text{odd}} (1 - P_K)\| \stackrel{?}{=} 1/3$ as a research-front placeholder (Chunk A §10 RF-1 entry, page 8 line 305; Chunk F §6.2(b) page 37 line 1523; Chunk H closing page 53 line 2219). RF-1 therefore introduces new definitional content (the algebraic decomposition into layer projectors), not pure consolidation. This is consistent with RF-1's research-increment framing (the RES suffix in RCF-RES-v6.1-RF1 denotes research increment), but the audit's patch P5 (see §6) is to flag this explicitly in §1.0 so that readers do not mistake "research increment" for "consolidation of existing definitions".

4.5 Bridging Identity in V6_0

V6_0 page 53 line 2224 carries the bridging identity $\|PQ(1 - K)\|^2 = \|QPQ(1 - K)\|$ and credits it to "the v5.1 audit" (page 8 line 306). The v5.1 audit document is not in the corpus; the attribution cannot be verified. However, the identity itself is the standard C*-algebra norm identity (valid for any bounded operators A , B with A self-adjoint and idempotent: $\|AB\|^2 = \|(AB)^T(AB)\| = \|B^T A^T AB\| = \|B^T A^T A B\| = \|BAB\|$ when $A = A^T = A^2$). The audit's patch P8 (see §6) is to re-cite the identity as a standard C*-algebra result, preserving the "v5.1 audit" attribution as a corpus-internal lineage note but not as the primary citation.

4.6 AALV / C-02 Citation Status in V6_0

V6_0's C-02 (AALV Theorem 1, Aharonov–Arad–Landau–Vazirani 2009 detectability lemma) carries a [CITE-VERIFY] flag (per V6_0 §6.2(b) DL-12, page 32 lines 1347–1352). The v5.1 corpus had cited a specific theorem number from AALV; the v5.1 audit retracted that citation as unverified; C-02 is currently name-level only (i.e., the citation acknowledges the AALV result exists but does not invoke a specific theorem form).

RF-1 sidesteps this issue cleanly: it invokes only the standard C^* -algebra B^*B identity, not the AALV theorem form. RF-1 does not formally "exhibit the detectability lemma"; it exhibits the bare operator-norm equality. The "Detectability Lemma" framing in RF-1's title is method-attribution, not theorem-form invocation. The audit's patch P9 (see §6) is to add a "Scope of Closure" note to §1 making this distinction explicit and confirming that the [CITE-VERIFY] flag on C-02 is preserved unchanged by RF-1.

5. Cross-Version Lineage Audit (v1 → v6)

This section traces the version chain $v1 \rightarrow v2 \rightarrow v2.1 \rightarrow v3 \rightarrow V4 \rightarrow V6$ to identify when RF-1’s specific content (AKLT chain, DL bound, sandwich operator, VN-17/VN-18, layer projectors) was introduced, and to flag any drift between versions that could affect RF-1’s interpretation. Agents B, C, and D read v3.0, V4_0, the Change Log, Patch Plan, Peer Review, Variance Retraction, and Con.pdf.

5.1 Version Chain Lineage

Transition	Date	Key changes
v1 → v2	2026-07-31	131 changes; 10→8 sections; STRIP-AND-DEFER acyclicity; Honesty Campaign (9 downgrades); SOE/MOE→Open Expansion; NEW Dark Energy §2.2; M_hat_red defined; Layer-Scheme A/B/C retired
v2 → v2.1	2026-08-02	15 patches (5 FATAL, 8 MATERIAL, 2 COSMETIC); 9 mandatory; 6 deferred; 8 validations preserved. F4 demotes T-4 CLOSED→STRENGTHENED; M4 mandates "calibrated" not "derived" for κ_B , $\Pi_{\max}$, $C=8\pi$
v2.1 → v3	Aug 2026	7 architectural changes: RP→Open Expansion reordering; Two-Link from Leibniz; F_MM replaces F; D=3 closed via SU(n) transitivity; 4 retractions (Universal Geometry Lemma, Layer-Closure Selector, Hodge-Duality, Sector-Dependent n); inflation-replacement added as 5th primitive
v3 → V4	post-Aug 2026	118pp expansion. Addresses Peer Review issues: Thm 1.4.1 Clifford Dichotomy (state-dependence for Clifford); §6.5 Strict FIREWALL Guardrail; Guardrail 4 demotes inflation-replacement → 4 primitives; Def 8.3.1 adds [Calibrated] category
V4 → V6	(most recent)	Introduces entire RF-1 increment: VN entry system (VN-01..VN-28, classes E/IV/C/F/O/L); 8-tag status system; §6.2(b) AKLT chain gap mainline; AKLT L=4 chain; sandwich operator (as RF-1 research Q); bridging identity; J-sector; AM-1/AM-3; DL-13; CDL-2; AALV/C-02 [CITE-VERIFY]

Table 5.1: Version chain $v1 \rightarrow V6$, with key changes per transition.

5.2 VN-17 / VN-18 Provenance

VN-17 and VN-18 are **introduced in V6_0**; they are absent from V4_0, v3.0, v2.1, v2.0, v1.0, and the Variance Retraction addendum. Grep for "VN-17", "VN-18", "VN-", "Verified-Numbers", "Class E", "Class IV", "Class O", "Class L", "Class F", "Class C" returned zero matches in V4_0 and all earlier versions. The status upgrade "Class O → Class IV / E [Established]" is therefore a V6_0-internal story: there is no v3.0 or V4_0 baseline against which to compare. RF-1’s upgrade is the first status change for VN-17/VN-18 since their introduction.

5.3 §6.2 Numbering Collision (V4_0 vs V6_0)

V4_0’s §6.2 is "Redundant Record Robustness / Quantum Darwinism" (same as v2.0/v2.1/v3.0’s §6.2 cosmology/records). V6_0’s §6.2(b) is "AKLT Frustration-Free Interacting Benchmark" — completely different content under the same section number. A reader cross-referencing V4_0 §6.2 and V6_0 §6.2(b) sees entirely different subjects. This is a numbering collision in V6_0’s renumbering (not introduced by RF-1), but RF-1 inherits the collision when it cites §6.2(b). The audit’s patch P6 (see §6) is to add a disambiguating note in §4.3, with the structural fix (renumbering §6.2(b) to §6.4 or §6.A) recommended for v6.2.

5.4 DL-Bound Lineage (2/3 → 1/6 Retraction)

V6_0 Appendix D.2 (page 45 line 1909) records: "DL bound: $2/3 \rightarrow 1/6$ (prefactor, v4.8 audit)". The v4.7 corpus carried $\Delta_{DL} = 2/3$; the v4.8 audit halved the prefactor to $1/6$. RF-1 silently inherits the post-retraction value without flagging the lineage. The "v4.8 audit" document is not in the corpus; the lineage is preserved as recorded by V6_0 and cannot be independently verified by this audit. The audit's patch P7 (see §6) is to add a footnote to §4.2 noting the retraction history and the unverifiability of the v4.8 audit document.

5.5 Peer Review Unresolved Issues

The Critical Peer Review (v2.1 vs v3.0, ~123 KB) raised 5 headline issues and identified two highest-priority mathematical results (Result A = Close T-2 for F_{MM} ; Result B = Derive inflation-replacement from RP minimization) that would convert MAJOR REVISION to MINOR REVISION. V4_0 partially addressed both (Theorem 1.4.1 Clifford Dichotomy for Clifford systems; Guardrail 4 primitive-count reduction), but the non-Clifford case and the full derivation remain open. The audit found that RF-1 inadvertently **worsens two Peer Review concerns**:

(a) **"Calibrated, not derived" pattern (Patch Plan M4, Peer Review issue iv)**. V6_0's VN-14/VN-15 ($\Delta_{\infty} \approx 0.350$, $\Delta_{std} > 0.592$) with White/Kennedy-Tasaki DMRG bulk-gap value (literature-derived, [CITE-VERIFY] flag) is the same "calibrated, not derived" issue Peer Review flagged for $\kappa_B/C=8\pi/S_{BH} 1/4$. RF-1's "every term derived from closed-form algebra or established finite-block theorems" wording understates this literature dependency. Patch P4 (see §6) addresses this.

(b) **"Status inflation" pattern (Patch Plan F4, Peer Review issue iv)**. V6_0's upgrade of VN-17/VN-18 to Class IV/E [Established] removes V4_0's honest OPEN candidate posture. The Peer Review critiqued this pattern in v3.0 (where T-4 was demoted from CLOSED to STRENGTHENED conditional on T-2). RF-1's reversal of DL-13 (re-inserting the DL term into the mainline as Established) is the same pattern. Patch P3 (see §6) addresses this by preserving the AM-3 dependency as a lineage annotation.

5.6 Variance Retraction Orthogonality

The Variance Retraction addendum (~25 KB) is a post-v3.0 scope correction that retracts the variance-form candidate B_{MM}^{var} and 9 over-extended cosmological claims. It does **not** touch v3.0's core algebraic theorems (it preserves $F_{MM} = \omega(J^2) \pi(\Delta) \dagger \pi(\Delta)$ per v3.0 Def. 1.17). It is orthogonal to AKLT/DL-bound content: grep for "AKLT", "Knabe", " γ_{DL} ", " Δ_{DL} ", " $1/6$ ", " $2/3$ ", "sandwich", "P_odd/P_even", "AALV", "C-02", "VN-17", "VN-18" returned zero matches in the addendum. The " $2/3$ " reference in the addendum is for $n_s = 2/3$ (cosmological scalar spectral index), unrelated to the DL bound. RF-1 is therefore not in conflict with the Variance Retraction addendum.

5.7 Con.pdf Orthogonality

Con.pdf (462 KB, 184 pages) is a standalone manuscript, **not** a compendium of the corpus. It has a simpler 4-category status taxonomy (Definition, Conditional Theorem, Open Theorem Target, Quarantined Speculation), no VN entry system, no AKLT chain, no DL bound, no sandwich operator, no bridging identity, no AALV/C-02 citation, no J-sector decomposition, no ψ_{top} . It represents an earlier or parallel lineage of the Relational Constraint Framework with different primitives (algebra + GNS, no Lie-Jordan decomposition). RF-1 cannot claim continuity with Con.pdf's content — it is an entirely new research layer relative to Con.pdf.

6. Findings — Severity-Tagged Issue List

This section lists all 14 findings from the audit, severity-tagged. The math is verified correct (no BLOCKER); all 14 issues are documentation, provenance, or consistency concerns. The redlined patch document at `/home/z/my-project/download/RCF-RES-v6.1-RF1-audit-patch.md` contains the full proposed text for each patch.

6.1 Summary Table

Patch	Severity	Title	Status
P1	MAJOR	Class/Status conflation in "Class IV / E [Established]"	Must fix
P2	MAJOR	§6.2(b) mainline truncation (ϵ -tail dropped)	Must fix
P3	MAJOR	§6.2(b) DL-term reinsertion reverses DL-13 gate	Must fix
P4	MAJOR	"Closed-form algebra" understates Δ_∞ Class L	Must fix
P5	MAJOR	P_{odd} / P_{even} are new definitions, not consolidation	Must fix
P6	MAJOR	§6.2 numbering collision (V4_0 vs V6_0)	Must fix
P7	MINOR	$\Delta_{\text{DL}} = 1/6$ lineage ($2/3 \rightarrow 1/6$ retraction) not flagged	Recommended
P8	MINOR	Bridging identity provenance unclear (v5.1 audit not in corpus)	Recommended
P9	MINOR	AALV / C-02 citation status not addressed	Recommended
P10	MINOR	Status tag history (v3 $\rightarrow$ V6 drift) not documented	Recommended
P11	MINOR	Mainline values rounded inconsistently	Recommended
P12	NIT	"[CLOSED — Proven in Exact Arithmetic]" tag informal	Optional
P13	NIT	"Theorem RF-1.1" numbering conflates register ID	Optional
P14	NIT	"Class O" vs "Class 0" ambiguity	Optional

Table 6.1: All 14 audit findings, severity-tagged.

6.2 MAJOR Findings (P1–P6) — Detailed Discussion

P1. Class/Status conflation

Location: RF-1 §1, §4.1.

Issue: RF-1 writes "Class IV / E [Established]" conflating VN provenance classes (V6_0 §6 closed set: E, IV, C, F, O, L) with §5 status tags (8 bracketed tags). V6_0's registry has a single Class column per VN entry, not multiple. The "IV / E" notation is ambiguous: it could mean "Class IV or Class E" or "Classes IV and E jointly". The slash is non-standard.

Evidence: V6_0 page 3 line 124 ("E exact · IV independently verified · C cited · F fitted · O open inference"); V6_0 Appendix A VN-17 entry (page 43 line 1828) uses a single Class column = "O".

Proposed patch: Restate as a two-column upgrade: *Class O* $\rightarrow$ *Class IV* (provenance); [*Input Verified*; *Inference OPEN*] $\rightarrow$ [*Established*] (status).

P2. §6.2(b) mainline truncation

Location: RF-1 §4.3.

Issue: RF-1's mainline stops at Δ_∞ ; V6_0's final frozen mainline extends to $\epsilon_6 \approx 0.398 \leq \epsilon_4 \approx 0.449 \leq \epsilon_3 = 0.500$. V6_0 calls this extended chain "static" (the ϵ_n sequence's asymptotic behavior is governed by the scaling law VN-13, not by this chain), but the chain still carries the local gaps as a reference. RF-1's truncation loses this

reference without explanation.

Evidence: V6_0 §6.2(b) page 37 lines 1527–1532 + DL-13 gate page 38 lines 1584–1590.

Proposed patch: Restore the ε -tail with a parenthetical clarifying that it is "static".

P3. DL-term reinsertion reverses DL-13

Location: RF-1 §4.3.

Issue: RF-1 re-inserts $\Delta_{DL} = 1/6$ into the main chain as a CLOSED Established bound. V6_0's DL-13 gate modification had explicitly removed the DL term from the main chain and presented it as a separately-stated OPEN candidate bound above the chain, with the VN-17/VN-18 dependency in prose only. This is a falsification safety mechanism that RF-1 reverses.

Evidence: V6_0 DL-13 (page 38, lines 1584–1590).

Proposed patch: Insert a "Note on DL-13 reversal" before the §4.3 mainline, noting that Theorem RF-1.1 closes VN-17 and justifies the reinsertion, and preserving AM-3 dependency as a lineage annotation (if Theorem RF-1.1 is later falsified, DL reverts to OPEN-candidate status per DL-13).

P4. "Closed-form algebra" understates Δ_{∞} Class L

Location: RF-1 §4.3 closing line.

Issue: V6_0's Δ_{∞} (VN-15) is anchored on $VN-14 = \Delta_{std} \approx 0.700$, which is Class L (literature-derived benchmark from White 1992 DMRG / Kennedy–Tasaki 1992), with a [CITE-VERIFY] flag. The robustness bound $\Delta_{std} > 0.592$ absorbs citation-risk, but the value is literature-derived, not closed-form algebra. RF-1's wording "every term is derived from closed-form algebra or established finite-block theorems" misrepresents this lineage. The Peer Review (v2.1 vs v3.0) issue (iv) flags exactly this pattern — "headline physics claims calibrated not derived" — and Patch Plan M4 mandates the "calibrated vs derived" distinction.

Evidence: V6_0 §6.2(b) page 35 lines 1498–1500; Appendix A page 43 line 1820.

Proposed patch: Replace with three-source classification: (i) closed-form algebra (Δ_{DL} via Theorem RF-1.1); (ii) finite-block theorems ($\Delta_{Knabe}^{(n=4/6)}, \varepsilon_n$); (iii) literature-derived benchmarks with explicit [CITE-VERIFY] provenance (Δ_{∞} via $VN-15 \leftarrow VN-14 = \Delta_{std} \approx 0.700$ [Class L]).

P5. P_{odd} / P_{even} are new definitions

Location: RF-1 §1.

Issue: V6_0 does **NOT** define P_{odd} or P_{even} algebraically. V6_0 only references the sandwich expression $\|P_{odd} P_{even} P_{odd} (1 - P_K)\| \approx 1/3$ as a research-front placeholder (3 locations: page 8 line 305, page 37 line 1523, page 53 line 2219). RF-1 introduces new definitional content (the algebraic decomposition into layer projectors), not pure consolidation. The wording "[CLOSED — Proven in Exact Arithmetic]" suggests closure of an existing open question, not introduction of new definitions.

Evidence: V6_0 page 8, 37, 53 (sandwich as placeholder only); Agents A, B, D all confirm no P_{odd}/P_{even} in V6_0/v3.0/V4_0/Variance Retraction.

Proposed patch: Add a "Definitions" subsection §1.0 explicitly flagging new vs. inherited: bond projectors inherited from V6_0 §6.2(b) (VN-22); P_K inherited from V6_0 §3 (Lemma 3.1.1); **layer projectors P_{odd} , P_{even} introduced by this research increment.**

P6. §6.2 numbering collision (V4_0 vs V6_0)

Location: RF-1 §4.3 (cites §6.2(b)).

Issue: V4_0's §6.2 is "Redundant Record Robustness / Quantum Darwinism" (cosmology/records). V6_0's §6.2(b) is "AKLT Frustration-Free Interacting Benchmark" — completely different content under the same section number. A reader cross-referencing V4_0 §6.2 and V6_0 §6.2(b) sees entirely different subjects. RF-1 inherits the collision when it cites §6.2(b).

Evidence: Agent D §1 V4_0 §6.2 = "Redundant Record Robustness / Quantum Darwinism"; V6_0 §6.2(b) = "AKLT Frustration-Free Interacting Benchmark".

Proposed patch: Option (a) add a disambiguating note: "§6.2(b) refers to V6_0's AKLT section; this is distinct from V4_0's §6.2 Quantum Darwinism." Option (b) structural fix: renumber V6.2 §6.2(b) to §6.4 or §6.A.

6.3 MINOR Findings (P7–P11) — Summary

P7. $\Delta_{DL} = 1/6$ lineage not flagged. The value $1/6$ inherits a retraction correction from $2/3$ (v4.7 corpus) by the v4.8 audit (document not in corpus). RF-1 silently inherits the post-retraction value. *Proposed patch:* add a lineage footnote to §4.2.

P8. Bridging identity provenance unclear. RF-1 invokes the C^* -algebra identity $\|AB\|^2 = \|BAB\|$. V6_0 page 53 credits this to "the v5.1 audit" — not in corpus. *Proposed patch:* re-cite as standard C^* -algebra result; preserve "v5.1 audit" as corpus-internal lineage note.

P9. AALV / C-02 citation status not addressed. RF-1's title invokes "Detectability Lemma" framing. C-02 carries [CITE-VERIFY] flag in V6_0. RF-1 sidesteps by invoking only the C^* -algebra $B*B$ identity. *Proposed patch:* add "Scope of Closure" note to §1; confirm [CITE-VERIFY] flag preserved unchanged.

P10. Status tag history (v3 $\rightarrow$ V6 drift) not documented. V6_0's 8-tag system is a recent evolution (v3 had 6 categories; V4 had 6 with [Calibrated] precursor). VN-17/VN-18 don't exist in v3/V4. *Proposed patch:* add lineage note to §4.

P11. Mainline values rounded inconsistently. RF-1 uses 4 decimal places (0.1734, 0.2481, 0.2958, 0.3500); V6_0 uses 3 (0.173, 0.248, 0.296, 0.350). Both are correct. *Proposed patch:* standardize to 4 decimal places throughout RF-1 and v6.2.

6.4 NIT Findings (P12–P14) — Summary

P12. "Status: [CLOSED — Proven in Exact Arithmetic]" is an informal coinage not in V6_0's closed tag set. *Proposed patch:* replace with [Established]; move "Proven in Exact Arithmetic" to a footnote.

P13. "Theorem RF-1.1" numbering conflates register ID (RF-1) with theorem number (1.1). *Proposed patch:* rename to "Theorem 1.1" or "Theorem A.1" within the RF-1 document.

P14. "Class O" vs "Class 0" visual ambiguity (letter O vs numeral 0). *Proposed patch:* add a glossary entry at first use.

7. Audit Verdict & Recommendations

The audit verdict is split: mathematical verification passes cleanly; documentation / provenance / lineage carries six MAJOR issues that must be addressed before RF-1 is promoted to v6.2. The audit's recommendation is to apply patches P1–P6 in a v6.1.1 interim release (or directly in v6.2), batch P7–P11 in a follow-up sweep, and treat P12–P14 as optional stylistic.

7.1 Mathematical Verdict

ALL 11 NUMERICAL CHECKS PASSED. RF-1 Theorem RF-1.1 is arithmetically sound. The spectrum, maximizing state, B^*B identity, ground-space orthogonality, J-sector decomposition, and Detectability Lemma bound are all reproduced exactly by the standalone verification script using sympy exact rational arithmetic. No BLOCKER issues identified. The verification script is delivered at `/home/z/my-project/download/aklt_verify.py` and can be re-executed at any time to confirm.

7.2 Documentation / Provenance / Lineage Verdict

6 MAJOR + 5 MINOR + 3 NIT issues identified. None are mathematical blockers; all are documentation, provenance, or consistency concerns. The most substantively debatable patch is P3 (DL-term reinsertion reversing DL-13), which involves a deliberate engineering choice (reversing a falsification safety mechanism) that is logically permitted by Theorem RF-1.1 but should be explicitly annotated. P4 (closed-form algebra understates Class L dependency) is the most material provenance issue: it directly engages the Peer Review's "calibrated vs derived" concern (Patch Plan M4).

7.3 Recommended Action Plan

Immediate (before v6.2 integration):

- Apply P1 (class/status conflation) to §1, §4.1, §4.2.
- Apply P2 (restore ε -tail) to §4.3.
- Apply P3 (DL-13 reversal note) to §4.3.
- Apply P4 (three-source classification) to §4.3.
- Apply P5 (definitions subsection) at top of §1.
- Apply P6 (§6.2 disambiguating note) to §4.3.

Follow-up sweep (within 2 weeks):

- Apply P7 (1/6 lineage footnote) to §4.2.
- Apply P8 (re-cite bridging identity) in §1.
- Apply P9 (Scope of Closure note) in §1.
- Apply P10 (status tag history) in §4.
- Apply P11 (decimal place standardization) throughout.

Optional stylistic (any time):

- Apply P12 (formal status tag).
- Apply P13 (theorem numbering).
- Apply P14 (Class O glossary).

7.4 Deliverables

The audit produces three standalone deliverables, all archived in /home/z/my-project/download/:

- **aklt_verify.py** — standalone Python verification script (sympy exact rational). Can be re-run at any time.
- **aklt_verify_output.txt** — full output log of the verification run (11 checks, all PASS, including the full spectrum and J-sector table).
- **RCF-RES-v6.1-RF1-audit-patch.md** — redlined patch document with proposed text for all 14 patches.
- **RCF-RES-v6.1-RF1-audit-report.pdf** — this deep-dive PDF audit report.
- **verification_results.json** — machine-readable verification results.

Appendix A. Verification Script Structure (Condensed)

The standalone verification script `aklt_verify.py` is structured in 14 sections. The full source is delivered at `/home/z/my-project/download/aklt_verify.py` (32 KB, 580 lines). The condensed structure is:

```
# Section 1 - Build the spin-1 single-site basis and tensor-product chain # ALL_STATES =
list of 81 4-tuples (m1, m2, m3, m4) with mi in {1, 0, -1} # STATE_TO_IDX = dict mapping
tuple -> index 0..80 # Section 2 - Build the spin-2 bond projector Pi on two spin-1 sites #
clebsch_two_spins_s1_to_j(j_target, m_target, m1, m2) # build_two_site_spin2_projector() ->
9x9 matrix PI2 # Sanity: PI2 eigenvalues = {0 (mult 4), 1 (mult 5)} ✓ # Section 3 - Lift Pi
to the full 81-dim Hilbert space on sites (i, i+1) # kron_general(*matrices) -> Kronecker
product # lift_bond_projector(bond_starts_at) -> 81x81 matrix # Build PI_12, PI_23, PI_34 ✓
# Section 4 - Build P_odd, P_even, and the AKLT Hamiltonian M_hat # P_odd = (I - PI_12) * (I
- PI_34) # P_even = I - PI_23 # M_hat = PI_12 + PI_23 + PI_34 # Sanity: all Pi idempotent &
self-adjoint ✓ # Section 5 - Build the sandwich operator S = P_odd P_even P_odd (1 - P_K) #
K_basis = M_hat.nullspace() # 4 vectors -> dim K = 4 ✓ # P_K = B * (B^T B)^{-1} * B^T #
orthogonal projector # Sanity: P_K idempotent, self-adjoint, rank 4 ✓ # S_op = P_odd *
P_even * P_odd * (I_81 - P_K) # Sanity: S_op self-adjoint ✓ # Section 6 - Verify RF-1 §2:
full spectrum by J-sectors # eigvals = S_op.eigenvals() # exact sympy Rational # Spectrum =
{0: 69, 1/36: 1, 1/6: 3, 11/36: 3, 1/3: 5} ✓ # lambda_max = 1/3, mult 5 -> ||S||_inf = 1/3 ✓
# Section 7 - Verify RF-1 §3: maximizing state |psi_top> # psi_top = |1,0,1,0> - |1,0,0,1> -
|0,1,1,0> + |0,1,0,1> # S_psi = S_op * psi_top # S_psi == (1/3) * psi_top ✓ # Section 8 -
Verify RF-1 §3.5: P_K |psi_top> = 0 # PK_psi = P_K * psi_top == 0 ✓ # Section 9 - Verify the
B*B identity (RF-1 §1 consequence) # B = P_even * P_odd * (I - P_K) # BstarB = B.T * B #
BstarB == S_op ✓ (uses commutativity + idempotence) # ||B||_inf = sqrt(lambda_max(B*B)) =
sqrt(1/3) = 1/sqrt(3) ✓ # Section 10 - Verify the J-sector decomposition (RF-1 §2) # Build
S1, S2, S3, S4 spin-1 operators (Sx, Sy, Sz on each site) # J2 = Jx^2 + Jy^2 + Jz^2 where
J_alpha = sum_i S^alpha_i # J^2 eigenvalue spectrum: J=0 -> 3 (mult), J=1 -> 18, J=2 -> 30, #
J=3 -> 21, J=4 -> 9. Sum = 81 ✓ # Radial dims (3,6,6,3,1) x mults (1,3,5,7,9) = 81 ✓ #
Section 11 - Verify the spin-2 bond projector action on |1,1>, |1,0>, |0,1>, |0,0> # Pi|1,1>
= |1,1> ✓ # Pi|1,0> = (1/2)(|1,0> + |0,1>) ✓ # Pi|0,1> = (1/2)(|1,0> + |0,1>) ✓ # Pi|0,0> =
(2/3)|0,0> + (1/3)(|1,-1> + |-1,1>) ✓ # Section 12 - Verify (1/8 + 1/12 + 1/8) = 1/3
arithmetic # Rational(1,8) + Rational(1,12) + Rational(1,8) = 1/3 ✓ # Section 13 - Verify
Detectability Lemma bound (VN-18) # gamma = 1/sqrt(3), g = 2 # Delta_DL = (1 - gamma^2) / g^2
= (1 - 1/3) / 4 = 1/6 ✓ # Section 14 - Summary # 11/11 checks PASSED # Results saved to
/home/z/my-project/audit/verification_results.json
```

Appendix B. Cross-Reference Table: RF-1 Claims vs Corpus

This appendix provides a per-claim cross-reference table showing where each of RF-1’s key claims is supported (or first introduced) in the corpus, and what status it carries in each version.

RF-1 claim	V6_0	V4_0	v3.0	Con.pdf
$\dim H = 81$	§6.2(b) p.35	absent	absent	absent
$\dim K = 4$	§6.2(b) p.35	absent	absent	absent
J-sector (3,6,6,3,1)x(1,3,5,7,9)	VN-06, Class IV	absent	absent	absent
$\Pi_{12}, \Pi_{23}, \Pi_{34}$	§6.2(b), VN-22	absent	absent	absent
$P_{\text{odd}}, P_{\text{even}}$	absent	absent	absent	absent
Sandwich op S	RF-1 placeholder, p.8/37/53	absent	absent	absent
$\ S\ = 1/3$	RF-1 placeholder $? = 1/3$	absent	absent	absent
$\gamma_{\text{DL}} = 1/\sqrt{3}$ (VN-17)	Class O, [IV OPEN]	absent	absent	absent
$\Delta_{\text{DL}} \geq 1/6$ (VN-18)	Class O, $\leftarrow$ VN-17	absent	absent	absent
B*B identity	p.53 line 2224 (v5.1)	absent	absent	absent
$\Delta_{\text{Knabe}}^{(n=4)} \approx 0.173$	VN-10, Class IV	absent	absent	absent
$\Delta_{\text{Knabe}}^{(n=6)} \approx 0.248$	VN-11, Class IV	absent	absent	absent
$E^* \approx 0.296$	VN-13, Class F	absent	absent	absent
$\Delta_{\infty} \approx 0.350$	VN-15, Class IV	absent	absent	absent
$\Delta_{\text{std}} \approx 0.700$	VN-14, Class L	absent	absent	absent
AALV / C-02	[CITE-VERIFY]	informal, Remark 1.4.1	§1.9 cited	absent
$ \psi_{\text{top}}\rangle$	absent	absent	absent	absent
$\Delta_{\text{DL}} = 2/3 \rightarrow 1/6$ lineage	App. D.2 line 1909	absent	absent	absent

Table B.1: Per-claim cross-reference across the corpus. "absent" = claim not present in that version.

The table confirms the version-chain story: the AKLT/DL-bound content is entirely a V6_0 introduction, with no v3.0, V4_0, or Con.pdf precedent. RF-1’s layer projectors $P_{\text{odd}} / P_{\text{even}}$ and maximizing state $|\psi_{\text{top}}\rangle$ are not present in any earlier version. The Δ_{DL} lineage ($2/3 \rightarrow 1/6$) is recorded in V6_0 Appendix D.2 only; the v4.8 audit document that performed the retraction is not in the corpus.